

Theoretical analysis of subwavelength high contrast grating reflectors

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Abstract: A simple analytic analysis of the ultra-high reflectivity feature of subwavelength dielectric gratings is developed. The phenomenon of ultra high reflectivity is explained to be a destructive interference effect between the two grating modes. Based on this phenomenon, a design algorithm for broadband grating mirrors is suggested.

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1. Introduction

Subwavelength gratings are of interest for a wide range of integrated optoelectronic device applications, including lasers, filters, splitters, couplers, etc., because the elimination of non-zero diffraction orders increases coupling efficiency. Recently, we proposed and demonstrated subwavelength dielectric gratings with a high contrast of refractive indices, referred to as high contrast gratings (HCGs), having reflectivity higher than 99% over an extraordinarily broad wavelength range of $\Delta\lambda/\lambda \sim 30\%$ [1,2]. Such high reflectivity is unexpected, since a uniform slab layer made of the same dielectric material (refractive index of 2.8~3.5) can only reach reflectivity of up to $\sim 70\%$. Highly reflective subwavelength

gratings were also experimentally shown to have a promising application in vertical cavity surface emitting lasers (VCSELs), in which they were monolithically integrated as a replacement of conventional distributed Bragg reflectors (DBRs) [2,3], and as a means to increase tuning speed in a tunable VCSEL [4] and to provide polarization control [5–7]. In addition, under a different design condition, HCG was shown to behave as an optical resonator with an ultra-high quality factor [8]. Due to the inhomogeneous refractive index profile, and due to the fact that the wavelength is comparable to the grating periodicity, the existing rigorous framework for the electromagnetic analysis of such gratings [9–11] is fairly complex, which makes it difficult to develop simple intuitive explanations for phenomena such as ultra-high broadband reflectivity, in a manner that will allow to *predict and design* this extraordinary and unexpected feature of HCGs. A formulation that combines rigor with simplicity remains yet to be presented.

In this work, we provide a straightforward, intuitive and yet *fully analytic* solution of HCGs, focusing on the high reflectivity phenomenon, without using either coarse rules of thumb or heavy mathematical formalisms, and provide a design algorithm for broadband highly-reflective HCGs. Special attention is paid to the multi-mode nature of such gratings, and to the very quick convergence of their modal representation.

2. Theoretical analysis of the grating reflectivity

In order to keep the analysis simple, we limit ourselves to the case of surface-normal incidence and a rectangular profile of refractive index. The grating geometry is described in Fig. 1a. The colored bars represent a dielectric material with a refractive index n_{bar} , which is significantly higher than the refractive index of the surrounding medium (hence the terminology “*high contrast grating*”). The typical refractive index of the grating bars is $n_{\text{bar}}=2.8\sim 3.5$, and the outside medium is assumed to be air ($n_{\text{air}}=1$), even though other low index media, such as silicon dioxide, produce comparable effects [1]. The notations Λ , a , s and t_g in Fig. 1a correspond respectively to the period of the grating, the air-gap size, the width of the grating bars and the grating thickness. The grating period is sub-wavelength ($\Lambda < \lambda$), but remains larger than λ/n_{bar} . The grating periodicity direction is x . The incident plane wave propagation direction, as indicated by the red arrow, is z . For simplicity, the grating is assumed to be infinite in y direction and infinitely periodic in x direction. We consider two polarizations of incidence: (i) transverse magnetic (TM), in which the electric field is in x direction; (ii) transverse electric (TE), in which the electric field is in y direction. Figure 1a describes both polarizations, the black arrows indicating the directions of the electric field in each case. Figure 1b shows an example of a broadband high reflectivity spectrum exhibited by such HCG.

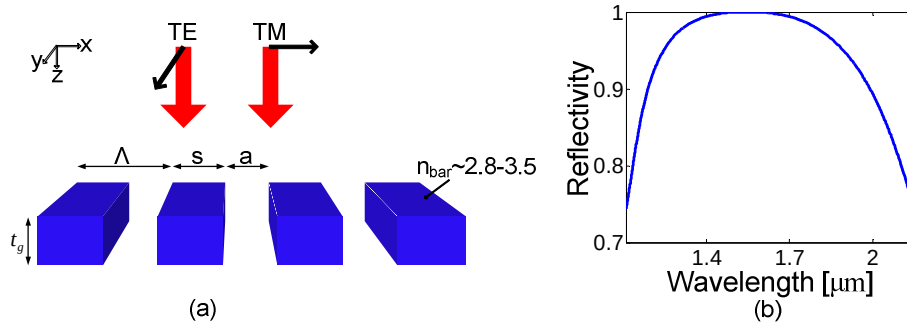


Fig. 1. (a) High Contrast Grating (HCG) schematics. The red arrow indicates the direction of wave incidence. The black arrows correspond to the E-field direction in both TE and TM polarizations of incidence. The grating comprises of rectangular dielectric bars having a refractive index in the semiconductor range, surrounded by a low index medium (air or oxide). The high refractive index contrast between the grating bars and the surrounding medium is beneficial for the reflectivity bandwidth and the fabrication tolerance of the grating. The nomenclature for the HCG dimensions is as follows: Λ is the grating period, a is the air-gap width, s is the bar width and t_g is the grating thickness. HCG has subwavelength periodicity ($\Lambda < \lambda$). (b) Example of the broadband high reflectivity spectrum. The HCG parameters for this plot are: $n_{\text{bar}}=3.21$, $\Lambda=0.779\mu\text{m}$, $s/\Lambda=0.77$, $t_g=0.508\mu\text{m}$ and TM polarization of incidence.

2.1 TM-polarized incidence

For simplicity, in this section we only focus on TM-polarized incidence, and in section 2.2 we list the differences for the TE case. In our solution, we consider three regions, separated by two planes: $z=-t_g$ (HCG input plane) and $z=0$ (HCG output plane), as shown in Fig. 2. We will keep track of the lateral (x,y) field components only, since the longitudinal (z) field component can be easily derived from the lateral components. Since the HCG is infinite in y , the solution is two-dimensional ($\partial/\partial y=0$). In region I, $z < -t_g$, there are incoming and reflected waves. In region II, $-t_g < z < 0$, the solutions are modes of a periodic array of slab waveguides, whereby the propagation direction is z . In region III, $z > 0$, there exist only the transmitted waves. We first solve for the modes in Region II, and then match the boundary conditions with Regions I and III to solve for reflectivity and transmission spectra.

The modes in *Region II* are given in Eq. (1a). They comprise of forward ($+z$) and backward ($-z$) propagating components, the coefficients of which are a_m and a_m^ρ respectively, $m=1,2,\dots$ being the number of the mode and β being the longitudinal (z) wavenumber. $h_{y,m}^{\text{in}}$ and $e_{x,m}^{\text{in}}$ in Eq. (1a) indicate the lateral (x) magnetic and electric field profiles of each mode, where the index “in” stands for: inside HCG.

$$\begin{aligned} H_y^{\text{II}}(x, -t_g \leq z \leq 0) &= \sum_{m=1}^{\infty} h_{y,m}^{\text{in}}(x) [a_m \exp(-j\beta_m z) - a_m^\rho \exp(+j\beta_m z)] \\ E_x^{\text{II}}(x, -t_g \leq z \leq 0) &= \sum_{m=1}^{\infty} e_{x,m}^{\text{in}}(x) [a_m \exp(-j\beta_m z) + a_m^\rho \exp(+j\beta_m z)] \end{aligned} \quad (1a)$$

Since our intention is to eventually use vectors and matrixes instead of summations, we will now define based on Eq. (1a) the coefficient vectors $\mathbf{a}$ and $\mathbf{a}^\rho$, and the reflection matrix $\mathbf{\rho}$, which relates between these two vectors:

$$\begin{aligned} \text{Coefficient vectors: } \mathbf{a} &\stackrel{\Delta}{=} (a_1 \quad a_2 \quad \dots)^T; & \mathbf{a}^\rho &\stackrel{\Delta}{=} (a_1^\rho \quad a_2^\rho \quad \dots)^T \\ \text{Reflection matrix } \mathbf{\rho} \text{ at the output plane } (z=0): & \mathbf{a}^\rho \equiv \mathbf{\rho} \mathbf{a} \end{aligned} \quad (1b)$$

It is also instructive at this point to define, based on Eq. (1a), the HCG propagation matrix $\mathbf{\phi}$, which is a diagonal matrix containing the accumulated phases of each HCG mode:

$$\Phi_{n,m} = \exp(-j\beta_m t_g) \text{ for } n = m \text{ and } 0 \text{ for } n \neq m \quad (1c)$$

The lateral field profiles, $h_{y,m}^{\text{in}}$ and $e_{x,m}^{\text{in}}$, in Eq. (1a), are given by:

$$\begin{aligned} h_{y,m}^{\text{in}} \left(\underbrace{0 < x < a}_{\text{inside air gaps}} \right) &= \cos(k_{s,m}s/2) \cos[k_{a,m}(x-a/2)]; & e_{x,m}^{\text{in}} &= (\beta_m/k_0) \eta h_{y,m}^{\text{in}} \\ h_{y,m}^{\text{in}} \left(\underbrace{a < x < \Lambda}_{\text{inside HCG bars}} \right) &= \cos(k_{a,m}a/2) \cos\{k_{s,m}[x-(a+\Lambda)/2]\}; & e_{x,m}^{\text{in}} &= (\beta_m/k_0) n_{\text{bar}}^{-2} \eta h_{y,m}^{\text{in}} \end{aligned} \quad (1d)$$

In Eq. (1d), $\eta=120\pi\Omega$ is the vacuum wave impedance; $k_0=2\pi/\lambda$; k_a and k_s are the lateral (x) wavenumbers inside air-gaps and grating bars, respectively (see Fig. 2). In addition, the solution must be periodic with respect to Λ , and therefore:

$$\begin{aligned} h_{y,m}^{\text{in}}(x < 0 \text{ or } x > \Lambda) &= h_{y,m}^{\text{in}}(x \text{ modulo } \Lambda) \\ e_{x,m}^{\text{in}}(x < 0 \text{ or } x > \Lambda) &= e_{x,m}^{\text{in}}(x \text{ modulo } \Lambda) \end{aligned} \quad (1e)$$

The nomenclature for Eqs. (1a)–(1e) is shown in Fig. 2:

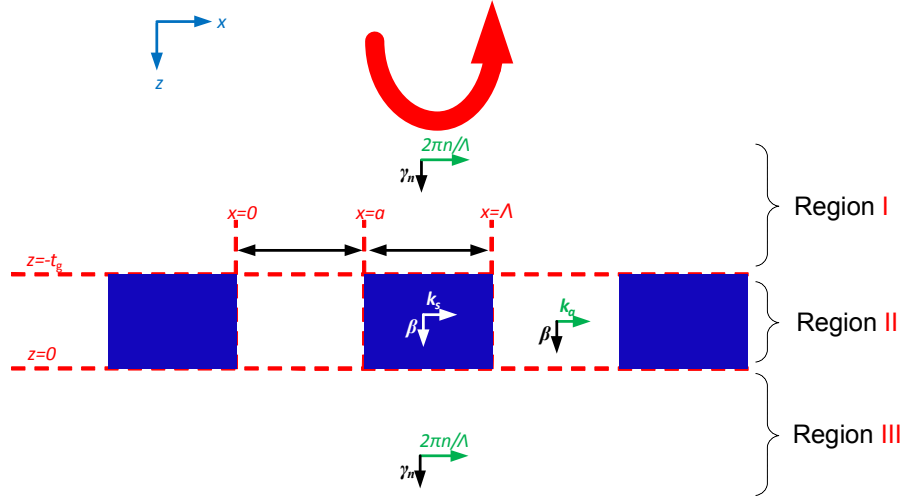


Fig. 2. Nomenclature for Eqs. (1a)–(1e): The HCG input plane is $z=-t_g$ and the output plane is $z=0$. k_a and k_s are the x -wavenumbers in the air-gaps and in the grating bars respectively. The z -wavenumber β is the same in both the air-gaps and the bars. The x -wavenumber (green arrow) outside the grating (Regions I and III) is determined by the grating periodicity: $2\pi n/\Lambda$.

The longitudinal wave number β in Eq. (1a) is given by:

$$\beta_m^2 = (2\pi/\lambda)^2 - k_{a,m}^2 = (2\pi n_{\text{bar}}/\lambda)^2 - k_{s,m}^2 \quad (1f)$$

Or alternatively:

$$k_{s,m}^2 - k_{a,m}^2 = (2\pi/\lambda)^2 (n_{\text{bar}}^2 - 1) \quad (1g)$$

From the perspective of the incident wave, the grating is merely a periodic array of (short) slab waveguides, whereby the propagation direction is along z . The dispersion relation between the lateral wavenumbers k_a and k_s therefore describes such array:

$$n_{\text{bar}}^{-2} k_{s,m} \tan(k_{s,m}s/2) = -k_{a,m} \tan(k_{a,m}a/2) \quad (2)$$

Equation (2) makes sure that the boundary conditions along $x=0$ and $x=a$ are matched for all field components. The characteristic Eq. (2) is presented in Fig. 3 for various grating duty

cycles (DC), defined as $DC=s/\Lambda$. The lower and upper curve-sets in Fig. 3 correspond to the first two solutions of Eq. (2), i.e. the first two modes, while the dashed lines in Fig. 3 are constant wavelength contours, presented in Eq. (1g). The intersections between the dashed lines and the curves, as marked by the black circles, indicate the modes at the specific wavelength. Figure 3 also shows the mode cutoff limit ($\beta=0$), which according to Eq. (1f) is given by $k_s=n_{\text{bar}}k_a$. Above the mode cutoff line, the HCG modes are propagating in z (real β) and below the mode cutoff line the modes are evanescent in z (imaginary β). Figure 3 shows that in HCG k_s is always real, while k_a can be either imaginary or real, depending on the wavelength. The lowest mode has only imaginary k_a values, and therefore its β has the largest value. Hence, we refer to the lowest mode as the fundamental, or the first, mode. This mode is also the only one not to have cutoff at large wavelengths. In fact, at large wavelengths ($\lambda \gg \Lambda$) the first mode resembles a plane wave ($k_s \sim 0$, $k_a \sim 0$). This is because at large wavelengths the exact grating corrugation profile loses effect, and the grating behavior approaches that of a uniform layer with an effective refractive index.

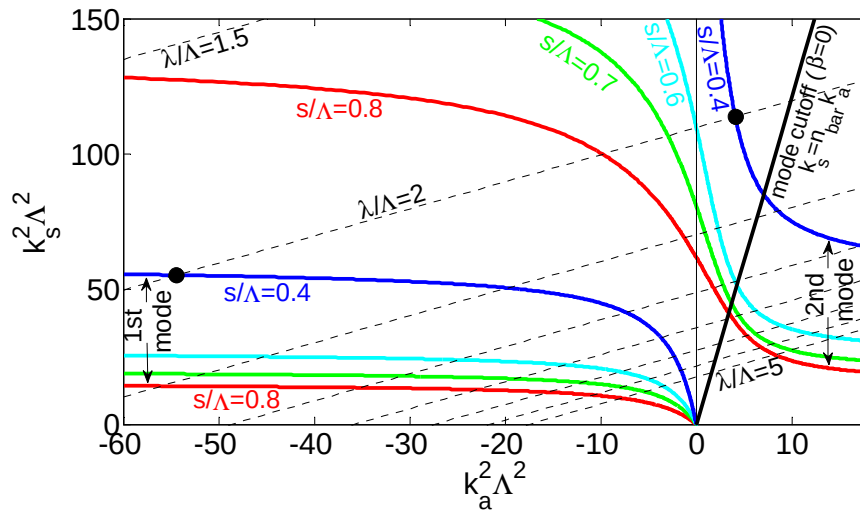


Fig. 3. Graphic representation of dispersion relations for a HCG with a refractive index $n_{\text{bar}}=3.48$, assuming TM polarization of incidence. Lower and upper curves are the first two solutions of Eq. (2), i.e. the first two HCG modes. Dashed lines are the constant wavelength contours, given by Eq. (1g). Black circles indicate the intersections between the dashed lines and the curves, thus describing the modes at a specific wavelength. The mode cutoff ($\beta=0$) is given by $k_s=n_{\text{bar}}k_a$, according to Eq. (1f). Above the cutoff β is real and below the cutoff β is imaginary.

The mode profiles in *Region I* are given by Eq. (3a). They comprise of an incident plane wave and multiple reflected modes (propagating in $-z$ direction), the coefficients of which are r_n , whereby $n=0,1,2,\dots$ is the number of the reflected mode. γ is the longitudinal ($-z$) wavenumber, and $h_{y,n}^{\text{out}}$ and $e_{x,n}^{\text{out}}$ in Eq. (3a) indicate the lateral (x) magnetic and electric field profiles, where the index “out” stands for: outside HCG.

$$\begin{aligned}
H_y^I(x, z \leq -t_g) &= \underbrace{\exp[-j(2\pi/\lambda)(z+t_g)]}_{\text{incident plane wave}} - \underbrace{\sum_{n=0}^{\infty} r_n h_{y,n}^{\text{out}}(x) \exp[+j\gamma_n(z+t_g)]}_{\text{reflected}} \\
&= \sum_{n=0}^{\infty} (\delta_{n,0} - r_n) h_{y,n}^{\text{out}}(x) \exp[+j\gamma_n(z+t_g)] \\
E_x^I(x, z \leq -t_g) &= \underbrace{\eta \exp[-j(2\pi/\lambda)(z+t_g)]}_{\text{incident plane wave}} + \underbrace{\sum_{n=0}^{\infty} r_n e_{x,n}^{\text{out}}(x) \exp[+j\gamma_n(z+t_g)]}_{\text{reflected}} = (3a) \\
&= \sum_{n=0}^{\infty} (\delta_{n,0} + r_n) e_{x,n}^{\text{out}}(x) \exp[+j\gamma_n(z+t_g)] \\
\delta_{n,0} &= \begin{cases} 1, & n = 0 \\ 0, & \text{otherwise} \end{cases}
\end{aligned}$$

$\delta_{n,0}$ in Eq. (3a) is known as the Kronecker delta function. We are now in the position to define the HCG reflectivity matrix $\mathbf{R}$, which relates between the incident wave coefficient, $\rho_{n,0}$, and the coefficients of the reflected modes r_n :

$$\text{HCG Reflectivity Matrix } \mathbf{R}: \quad (r_0 \quad r_1 \quad r_2 \quad \dots)^T \equiv \mathbf{R} \underbrace{(1 \quad 0 \quad 0 \quad \dots)^T}_{\delta_{n,0}} \quad (3b)$$

Equation (3c) presents the lateral (x) field profiles, $h_{y,n}^{\text{out}}$ and $e_{x,n}^{\text{out}}$, in region I (obviously region III will have the same lateral field profiles):

$$h_{y,n}^{\text{out}} = \cos[(2n\pi/\Lambda)(x-a/2)]; \quad e_{x,n}^{\text{out}} = (\gamma_n/k_0) \eta h_{y,n}^{\text{out}} \quad (3c)$$

Equations (3c) shows that each air-gap center ($x=a/2$) is a symmetry plane for all modes in region I. However, each grating bar center (e.g. $x=(a+\Lambda)/2$) is a symmetry plane as well. This is because each grating bar center is located half-period away from the adjacent air-gap centers. This is of course also true for the region-II lateral profiles $h_{y,m}^{\text{in}}$ and $e_{x,m}^{\text{in}}$ described in Eq. (1d). In addition, the fact that the plane wave incidence is surface normal means that the solution above has no preferred direction among $+x$ and $-x$, and therefore the modes in Eqs. (1), (3) have a standing wave (cosine) lateral profile. The lateral symmetry in Eqs. (1), (3) is even (cosine) rather than odd (sine), because the incident plane wave (Eq. (3a)) has a laterally constant profile, and thus it can only excite laterally-even modes.

The transmitted mode profiles for Region III are given by Eq. (3d):

$$\begin{aligned}
H_y^{\text{III}}(x, z \geq 0) &= \sum_{n=0}^{\infty} \tau_n h_{y,n}^{\text{out}}(x) \exp(-j\gamma_n z) \\
E_x^{\text{III}}(x, z \geq 0) &= \sum_{n=0}^{\infty} \tau_n e_{x,n}^{\text{out}}(x) \exp(-j\gamma_n z)
\end{aligned} \quad (3d)$$

Based on Eq. (3d), we can now define the transmitted coefficient vector $\boldsymbol{\tau}$ for the modes in region III, as well as the HCG transmission matrix $\mathbf{T}$, which relates between the incident wave coefficient, $\rho_{n,0}$, and the coefficients of the transmitted modes τ_n :

$$\begin{aligned}
\text{Transmitted coefficient vector: } \quad \boldsymbol{\tau} &= (\tau_0 \quad \tau_1 \quad \dots)^T \\
\text{HCG Transmission Matrix } \mathbf{T}: \quad (\tau_0 \quad \tau_1 \quad \tau_2 \quad \dots)^T &\equiv \mathbf{T} \underbrace{(1 \quad 0 \quad 0 \quad \dots)^T}_{\delta_{n,0}} \quad (3e)
\end{aligned}$$

The longitudinal (z) wavenumber γ in Eqs. (3a) and (3d) is given by:

$$\gamma_n^2 = (2\pi/\lambda)^2 - (2n\pi/\Lambda)^2 \quad (3f)$$

As evident from Eq. (3f), in subwavelength gratings ($\Lambda < \lambda$) only the 0th diffraction order is propagating (γ_0 is real), while the first, second and higher orders are all evanescent (γ_1, γ_2 etc. are imaginary).

Based on the mode profiles in Eqs. (1)–(3), we can now describe the calculation of the HCG reflectivity. We first match the boundary conditions at the HCG output plane ($z=0$). We start from the magnetic field H_y :

$$\underbrace{\sum_{n=0}^{\infty} \tau_n h_{y,n}^{\text{out}}(x)}_{=H_y^{\text{III}}(x,z=0), \text{eq. 3d}} = \underbrace{\sum_{m=1}^{\infty} h_{y,m}^{\text{in}}(x)(a_m - a_m^{\rho})}_{=H_y^{\text{II}}(x,z=0), \text{eq. 1a}} \quad (4a)$$

By performing a Fourier overlap integral on both sides of Eq. (4a) we can eliminate the summation at the left-hand-side and express the transmitted coefficients τ_n in terms of the overlaps between the lateral (x) magnetic field profiles $h_{y,m}^{\text{in}}$ and $h_{y,n}^{\text{out}}$:

$$\underbrace{\tau_n \Lambda^{-1} \int_0^{\Lambda} [h_{y,n}^{\text{out}}(x)]^2 dx}_{=(2-\delta_{n,0})^{-1}, \text{see eq. 3c}} = \sum_{m=1}^{\infty} (a_m - a_m^{\rho}) \Lambda^{-1} \int_0^{\Lambda} h_{y,m}^{\text{in}}(x) h_{y,n}^{\text{out}}(x) dx \quad (4b)$$

By performing the overlap integrals shown in Eq. (4b), we thus project the orthogonal set $h_{y,m}^{\text{in}}$ onto the orthogonal set $h_{y,n}^{\text{out}}$. The expression $(2-\delta_{n,0})$ in Eqs. (4b) accounts for the general fact in Fourier theory - that 0th self-overlap is always twice larger than all the subsequent (cosine) self-overlaps. By repeating the steps in Eqs. (4a), (4b), this time for the electric field E_x , we can now express the same transmitted coefficients τ_n in terms of the overlaps between the lateral *electric* field profiles:

$$\underbrace{\tau_n \Lambda^{-1} \int_0^{\Lambda} [e_{x,n}^{\text{out}}(x)]^2 dx}_{=(\eta\gamma_n/k_0)^2 (2-\delta_{n,0})^{-1}, \text{see eq. 3c}} = \sum_{m=1}^{\infty} (a_m + a_m^{\rho}) \Lambda^{-1} \int_0^{\Lambda} e_{x,m}^{\text{in}}(x) e_{x,n}^{\text{out}}(x) dx \quad (4c)$$

We can now finally make a transition from summations into vectors and matrixes. We do this by defining the overlap matrixes $\mathbf{H}$ and $\mathbf{E}$ (both are unit-less), for the magnetic and electric field profiles respectively, based on Eqs. (4b) and (4c), and then rewriting Eqs. (4b) and (4c) in matrix-vector format:

$$\begin{aligned} \mathbf{H}_{n,m} &= \Lambda^{-1} (2 - \delta_{n,0}) \int_0^{\Lambda} h_{y,m}^{\text{in}}(x) h_{y,n}^{\text{out}}(x) dx, & \text{see eq. 4b} \\ \mathbf{E}_{n,m} &= \Lambda^{-1} (2 - \delta_{n,0}) (\eta\gamma_n/k_0)^{-2} \int_0^{\Lambda} e_{x,m}^{\text{in}}(x) e_{x,n}^{\text{out}}(x) dx, & \text{see eq. 4c} \end{aligned} \quad (5a)$$

$$\text{Eqs. 4b,c in matrix-vector format: } \boldsymbol{\tau} = \mathbf{H}(\mathbf{a} - \mathbf{a}^{\rho}) = \mathbf{E}(\mathbf{a} + \mathbf{a}^{\rho})$$

Equation (5a) can be rewritten as follows, using the definition of $\boldsymbol{\rho}$ -matrix in Eq. (1b):

$$\boldsymbol{\tau} = \mathbf{H}(\mathbf{I} - \boldsymbol{\rho})\mathbf{a} = \mathbf{E}(\mathbf{I} + \boldsymbol{\rho})\mathbf{a} \quad (5b)$$

Since Eq. (5b) holds for every coefficient vector $\mathbf{a}$ we can now derive the reflection matrix $\boldsymbol{\rho}$ as a function of the overlap matrixes $\mathbf{E}$ and $\mathbf{H}$:

$$\mathbf{H}(\mathbf{I} - \boldsymbol{\rho}) = \mathbf{E}(\mathbf{I} + \boldsymbol{\rho}) \Rightarrow \boldsymbol{\rho} = (\mathbf{I} + \mathbf{H}^{-1}\mathbf{E})^{-1} (\mathbf{I} - \mathbf{H}^{-1}\mathbf{E}) \quad (6)$$

The reflection matrix $\mathbf{\rho}$ in Eq. (6) is typically non-diagonal, which means that the HCG modes in Region II couple into each other during the reflection. This does not contradict the orthogonality of the modes in region II, since the reflection involves interaction with the external modes of region III, which are not orthogonal to the modes in region II.

Having matched the boundary conditions at the HCG output plane ($z=0$) we now repeat the steps in Eqs. (4)–(6) in order to match the boundary conditions at the HCG *input* plane ($z=-t_g$). For simplicity, we this time omit the details shown in Eqs. (4)–(6) and jump straight to the final equation:

$$H_y^I(x, z = -t_g) = H_y^{II}(x, z = -t_g) \quad \text{and} \quad E_x^I(x, z = -t_g) = E_x^{II}(x, z = -t_g) \quad (7a)$$

$$\Rightarrow \text{same steps as in eqs. 4-6} \Rightarrow (\mathbf{I} - \mathbf{R})^{-1} \mathbf{H}(\boldsymbol{\varphi}^{-1} - \boldsymbol{\rho}\boldsymbol{\varphi}) = (\mathbf{I} + \mathbf{R})^{-1} \mathbf{E}(\boldsymbol{\varphi}^{-1} + \boldsymbol{\rho}\boldsymbol{\varphi})$$

By rearranging Eq. (7a), we can implicitly express the HCG reflectivity matrix $\mathbf{R}$ in terms of the matrixes $\mathbf{E}$, $\mathbf{H}$, $\boldsymbol{\rho}$ and $\boldsymbol{\varphi}$:

$$\mathbf{E}(\mathbf{I} + \boldsymbol{\rho}\boldsymbol{\varphi}) (\mathbf{I} - \boldsymbol{\rho}\boldsymbol{\varphi})^{-1} \mathbf{H}^{-1} = (\mathbf{I} + \mathbf{R}) (\mathbf{I} - \mathbf{R})^{-1} \stackrel{\Delta}{=} \mathbf{Z}_{in} \quad (7b)$$

A reader familiar with transmission lines [12] might recognize from Eq. (7b) the definition of the HCG normalized input impedance matrix $\mathbf{Z}_{in}$, which is the equivalent of the corresponding scalar in regular transmission line theory. Using Eq. (7b), the HCG reflectivity matrix $\mathbf{R}$ can finally be calculated, resulting in an equation very common in transmission line theory:

$$\mathbf{R} = (\mathbf{Z}_{in} + \mathbf{I})^{-1} (\mathbf{Z}_{in} - \mathbf{I}), \text{ where } \mathbf{Z}_{in} = \mathbf{E}(\mathbf{I} + \boldsymbol{\rho}\boldsymbol{\varphi}) (\mathbf{I} - \boldsymbol{\rho}\boldsymbol{\varphi})^{-1} \mathbf{H}^{-1} \quad (7c)$$

Having calculated the HCG reflection matrix $\mathbf{R}$, we now calculate the HCG *transmission* matrix $\mathbf{T}$. We first derive the coefficient vector $\mathbf{a}$ in terms of the matrixes $\mathbf{E}$, $\mathbf{H}$, $\boldsymbol{\rho}$ and $\boldsymbol{\varphi}$, using steps similar to Eqs. (4)–(6):

$$\mathbf{a} = 2\boldsymbol{\varphi} \mathbf{I} (\mathbf{Z}_{in}^{-1} + \mathbf{I}) \mathbf{E} (\mathbf{I} + \boldsymbol{\rho}\boldsymbol{\varphi}) \mathbf{I}^{-1} \underbrace{(1 \ 0 \ 0 \ \dots)^T}_{\delta_{n,0}} \quad (8a)$$

We then show the derivation of the transmission vector $\boldsymbol{\tau}$, from which the HCG transmission matrix $\mathbf{T}$ naturally emerges:

$$\boldsymbol{\tau} = \underset{\text{eq. 5b}}{\mathbf{E}(\mathbf{I} + \boldsymbol{\rho})} \underset{\text{eq. 8a}}{\mathbf{a}} = 2\mathbf{E}(\mathbf{I} + \boldsymbol{\rho}) \boldsymbol{\varphi} \mathbf{I} (\mathbf{Z}_{in}^{-1} + \mathbf{I}) \mathbf{E} (\mathbf{I} + \boldsymbol{\rho}\boldsymbol{\varphi}) \mathbf{I}^{-1} (1 \ 0 \ 0 \ \dots)^T \quad (8b)$$

$\stackrel{\Delta}{=} \mathbf{T}, \text{ HCG Transmission Matrix}$

Lastly, the HCG reflectivity and transmission, and the relation between the two in the subwavelength regime, are summarized in Eq. (9):

$$\begin{aligned} \text{HCG Reflectivity} &= |\mathbf{R}_{00}|^2 \\ \text{HCG Transmission} &= |\mathbf{T}_{00}|^2 \\ |\mathbf{R}_{00}|^2 + |\mathbf{T}_{00}|^2 &\equiv 1 \text{ when } \Lambda < \lambda \end{aligned} \quad (9)$$

The last relation in Eq. (9) only applies to subwavelength gratings, because such gratings do not have diffraction orders other than the zeroth (in this case – surface normal) order (see Eq. (3f)), and thus all power that is not transmitted through the zeroth diffraction order gets reflected back. This fact is essential for the design of high reflectivity gratings, since high reflectivity can be achieved in such gratings by merely cancelling the 0th transmissive diffraction order (i.e. $\tau_0=0$). Had there been more than one transmitted and reflected diffraction orders (i.e. when $\Lambda > \lambda$), obtaining high reflectivity ($|r_0| \sim 1$) through cancellation of multiple orders ($\tau_0=0$, $\tau_1=0$, $r_1=0$, etc...) would be very difficult, which is why ordinary diffraction gratings are typically *not* associated with ultra-high reflectivity phenomena.

2.2 TE-polarized incidence

The analysis for TE polarized incidence follows the same steps as in section 2.1. The only differences are summarized in Table 1. The reason for differences is simple: TM solution relies largely on Maxwell-Ampère's equation: $\nabla \times \vec{H} = j\omega\epsilon_0\epsilon_r\vec{E}$, whereas TE solution relies largely on Maxwell-Faraday's equation: $\nabla \times \vec{E} = -j\omega\mu\vec{H}$. The lack of ϵ_r in Faraday's equation explains why n_{bar}^{-2} is replaced by 1, and the minus sign in Faraday's equation explains the minus sign in the last two entries of Table 1. The inversion of the wavenumber-ratios at the last two entries of Table 1 is also a common difference between TM and TE modes in waveguides.

Table 1. Differences between TM and TE polarizations of incidence. Check-marks indicate the changes required for each equation. Equations not listed in this table require no changes

Change from TM to TE: Replace (1) by (2)		Equation:											
(1)	(2)	1a	1d	1e	2	3a	3c	3d	4a	4b	4c	5a	7a
H_y	H_x	✓				✓		✓	✓				✓
E_x	E_y	✓				✓		✓					✓
h_y	h_x	✓	✓	✓		✓	✓	✓	✓	✓		✓	
e_x	e_y	✓	✓	✓		✓	✓	✓			✓	✓	
n_{bar}^{-2}	1		✓		✓								
β_m / k_0	$-k_0 / \beta_m$		✓										
γ_n / k_0	$-k_0 / \gamma_n$						✓				✓	✓	

2.3 Solution convergence

The next step is to determine how many modes are actually required to obtain good precision, i.e. how quickly the solution in sections 2.1-2.2 converges. Figure 4a shows a convergence example for the TM polarization. For comparison, Fig. 4a also presents the HCG reflectivity calculated using Rigorous Coupled Wave Analysis (RCWA) [9]. Figure 4a demonstrates very good agreement between the analytic solution above and the RCWA simulation, especially when the reflectivity is high. A clear conclusion from Fig. 4a is that when the reflectivity is high, considering only the first two modes is already sufficient to describe the reflectivity with very good precision, which means that solution convergence is very fast. This confirms the underlying principle of the next sections, which is the *double-mode* nature of the HCG high reflectivity phenomenon. The fact that taking only two modes into account is a good approximation is a major advantage of the solution method described in sections 2.1-2.2. Such fast convergence is unlike the RCWA solution method, which is based on lateral (x) Fourier expansion of the permittivity ϵ_r , and thus requires a significantly larger number of modes in cases of a rectangular profile of refractive index, as considered here. Figure 4a also presents the difference (i.e. the error) between the RCWA solution and the solution described above, showing that when the reflectivity is high, the error between a double-mode solution and the RCWA solution is negligible.

The fact that taking only two modes into account is a good enough approximation is further validated in Fig. 4b, where the >99% reflectivity contours (red) are plotted as a function of wavelength λ and thickness t_g (both normalized by λ). Figure 4b shows that almost all high reflectivity configurations are concentrated between the cutoffs of the second and the third modes – a region where only the first two modes are propagating (β_1 and β_2 are real) and the third mode is below cutoff (i.e. β_3 is imaginary). High reflectivity in a triple-mode region is shown to be possible but rare, since the wavelengths of the triple-mode region

are too close to the subwavelength limit. *Broadband* high reflectivity (i.e. broad red contour sections, such as the one indicated by the red arrow) are non-existent in the triple-mode region. Notably, there are also no high-reflectivity contours in either the single-mode region or below the subwavelength limit.

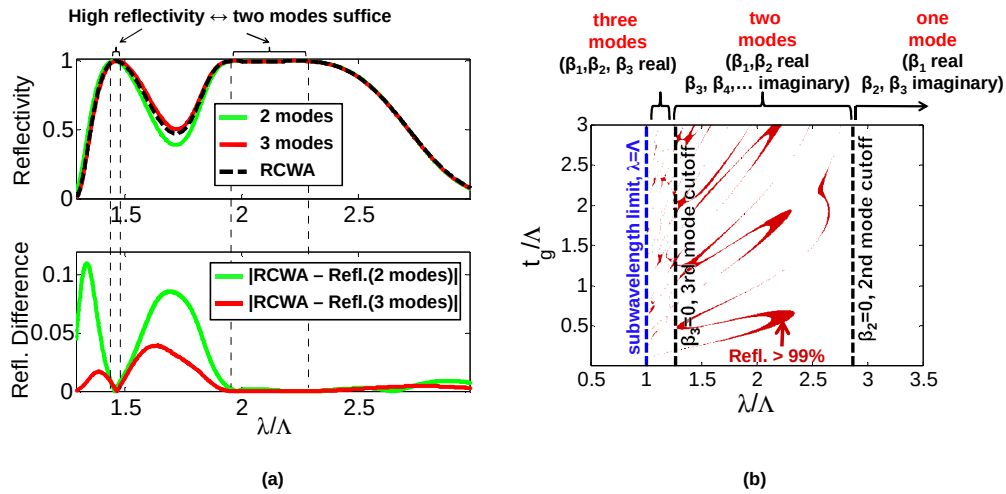


Fig. 4. (a) Convergence of the analytical solution in sections 2.1-2.2 towards the RCWA [9] simulation result as a function of the number of modes taken into consideration. The double-mode solution is in very good agreement with RCWA, especially when the reflectivity is high. The HCG parameters for this plot are: $n_{\text{bar}}=3.214$, $t_g/\Lambda=0.627$, $s/\Lambda=0.62$ and TM polarization of incidence. (b) $>99\%$ reflectivity contour as a function of wavelength λ and thickness t_g . Almost all high-reflectivity configurations are shown to reside within the double-mode region, i.e. between the cutoffs of the second and third modes.

A further examination of solution convergence is shown in Fig. 5, for TE polarization of incidence. Figures 5a–5d present the E-field intensity contours corresponding to $\sim 100\%$ reflectivity as a function of the number of modes taken into account (varying from 1 to 4). In a single-mode solution (Fig. 5a), the boundary condition cannot be satisfied, as seen by the intensity discontinuity at the HCG input and output planes. The double-mode solution in Fig. 5b, however, is already close to the final result. Figures 5e–5h decompose the overall intensity profile inside the HCG into the individual contributions of the first four modes. The first two modes have comparable intensities. Unlike the first two modes, the third mode is below cutoff, taking a form of a surface wave, evanescently decaying along z with the intensity ~ 25 times lower than the second mode.

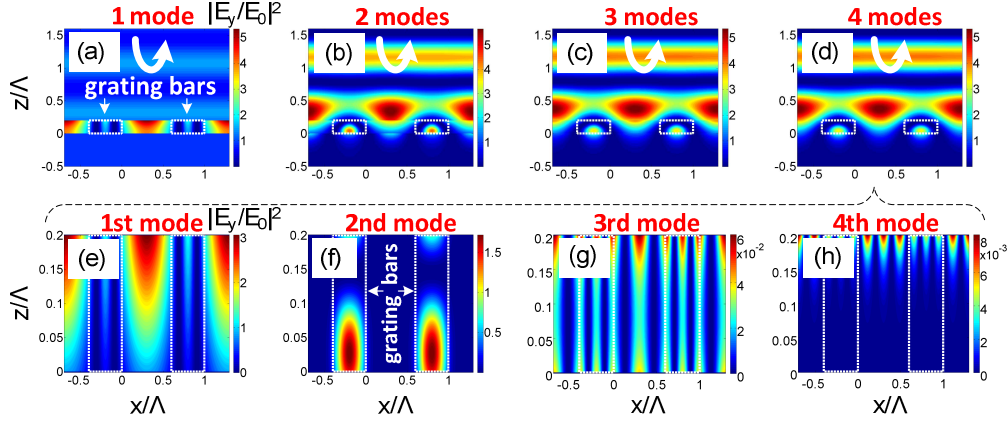


Fig. 5. (a-d) Convergence of the intensity profiles as a function of the number of modes taken into consideration, corresponding to the case of $\sim 100\%$ reflectivity. The grating bars are marked by the white dashed squares. When three or more modes are used, the boundary condition matching is almost perfect, but two modes are already a good approximation. The high reflectivity is clearly demonstrated by the standing wave profile created above the grating by the incident and reflected plane waves. (e-h) Contour plots of each mode separately inside the HCG. The grating bars are marked by the white dashed squares. The contours take into account both the forward (+z) and the backward (-z) propagating components of each mode, including their coefficients. The HCG parameters for this plot are: $n_{\text{bar}}=3.48$, $\lambda/\Lambda=1.509$, $t_g/\Lambda=0.2$, $s/\Lambda=0.4$ and TE polarization of incidence.

3. The mechanism of 100% reflectivity

Both Figs. 4 and 5 show that HCG is inherently a double-mode device, notwithstanding the additional small perturbation caused by the higher sub-cutoff evanescent modes. Therefore, intuitively, a 100% reflectivity phenomenon in HCGs is a *double mode* effect, the nature of which is examined in this section.

As established at the end of section 2.1, the condition for 100% reflectivity in subwavelength gratings is $\tau_0=0$. According to Eq. (5a), $\tau=\mathbf{E}(\mathbf{a}+\mathbf{a}^p)$, and therefore we can summarize the 100% reflectivity condition as follows:

$$|\mathbf{R}_{00}| = 100\% \leftrightarrow \tau_0 = [\mathbf{E}(\mathbf{a} + \mathbf{a}^p)]_0 = \sum_m \mathbf{E}_{0m} (a_m + a_m^\rho) = 0 \quad (10)$$

By using Eqs. (5a) and (3c), τ_0 can be rewritten as follows:

$$\begin{aligned} \tau_0 &= \sum_m \mathbf{E}_{0m} (a_m + a_m^\rho) \stackrel{\text{eq. 5a}}{=} (\eta\gamma_0/k_0)^{-2} \sum_m (a_m + a_m^\rho) \Lambda^{-1} \int_0^\Lambda e_{x,m}^{\text{in}}(x) e_{x,0}^{\text{out}}(x) dx \\ &\stackrel{\text{eq. 3c}}{=} (\eta\gamma_0/k_0)^{-1} \sum_m (a_m + a_m^\rho) \Lambda^{-1} \int_0^\Lambda e_{x,m}^{\text{in}}(x) dx = 0 \end{aligned} \quad (11)$$

Finally, by invoking a double-mode approximation, we obtain the simplified condition for 100% reflectivity, based on Eq. (11):

$$|\mathbf{R}_{00}| = 100\% \leftrightarrow \underbrace{(a_1 + a_1^\rho) \Lambda^{-1} \int_0^\Lambda e_{x,1}^{\text{in}}(x) dx}_{\text{lateral average of the first mode (forward + backward)}} + \underbrace{(a_2 + a_2^\rho) \Lambda^{-1} \int_0^\Lambda e_{x,2}^{\text{in}}(x) dx}_{\text{lateral average of the second mode (forward + backward)}} = 0 \quad (12)$$

"destructive interference" (cancellation) between the first and the second modes at $z=0^-$

Rephrasing Eq. (12), we see that 100% reflectivity is obtained and the zeroth transmissive diffraction order is suppressed ($\tau_0=0$) when the lateral average of the first mode and that of the

second mode cancel each other at the HCG output plane ($z=0$). We refer to such cancellation as “destructive interference”. We use quote-marks due to the unusual *lateral-average* interpretation of interference. The lateral average emerges from the overlap integral $\int e_{x,m}^{\text{in}} e_{x,0}^{\text{out}} dx$ in Eq. (11), as a consequence of $e_{x,0}^{\text{out}}$ being constant with respect to x . More generally, if we represent the overall field profile (from all grating modes combined) at the output plane $z=0$ in terms of a Fourier series as a function of x (Λ being the Fourier series periodicity), the lateral average will merely have the mathematical meaning of zeroth Fourier coefficient (“dc”-coefficient). It is easy to show that in general, n th Fourier coefficient in such expansion would correspond directly to the transmission coefficient τ_n of $\pm n$ th transmissive diffraction orders. However, since in subwavelength gratings only the zeroth order carries power along z (as mentioned above), it is only the “dc” Fourier coefficient that we need to suppress, as shown in Eq. (12). Moreover, since the intensities of the first two modes are comparable, as shown in Figs. 5e, 5f, achieving destructive interference between them is fairly straightforward, by merely adjusting the optical path phases of the modes which are determined by HCG thickness (t_g).

Figure 6a illustrates the destructive interference concept. The average lateral e-fields of the first two modes, $|(a_1 + a^{\rho}_1)A^{-1} \int e_{x,1}^{\text{in}} dx|$ and $|(a_2 + a^{\rho}_2)A^{-1} \int e_{x,2}^{\text{in}} dx|$ respectively (see Eq. (12)), are plotted along with their phase difference $\Delta\phi = \text{phase}[(a_1 + a^{\rho}_1)A^{-1} \int e_{x,1}^{\text{in}} dx] - \text{phase}[(a_2 + a^{\rho}_2)A^{-1} \int e_{x,2}^{\text{in}} dx]$. At the points of 100% reflectivity the modes are at anti-phase ($\Delta\phi = \pi$) with equal intensities, which means that perfect cancellation occurs (Eq. (12)). If two such 100% reflectivity points are located at close spectral vicinity, a broad-band of high reflectivity is achieved, as shown in Fig. 6a (top). Figure 6b illustrates the non-traditional “dc”-component interpretation for destructive interference: The lateral field profile (black curve, right plot), given by $(a_1 + a^{\rho}_1)E_{y,1}^{\text{II}}(x, z=0) + (a_2 + a^{\rho}_2)E_{y,2}^{\text{II}}(x, z=0)$, is plotted as a function of x for the case of perfect cancellation, showing that the field profile is zero only in terms of dc-component, but non zero otherwise. The individual field profiles of the first and the second modes, $(a_1 + a^{\rho}_1)E_{y,1}^{\text{II}}(x, z=0)$ and $(a_2 + a^{\rho}_2)E_{y,2}^{\text{II}}(x, z=0)$ respectively, are also plotted in Fig. 6b (blue and red curves, left plot). The “dc” components of the first two modes are shown to cancel each other. Had the grating not been subwavelength, this cancellation would no longer be enough, since in order to cancel higher diffraction orders, higher Fourier components (as opposed to only “dc”) would have to be zero as well.

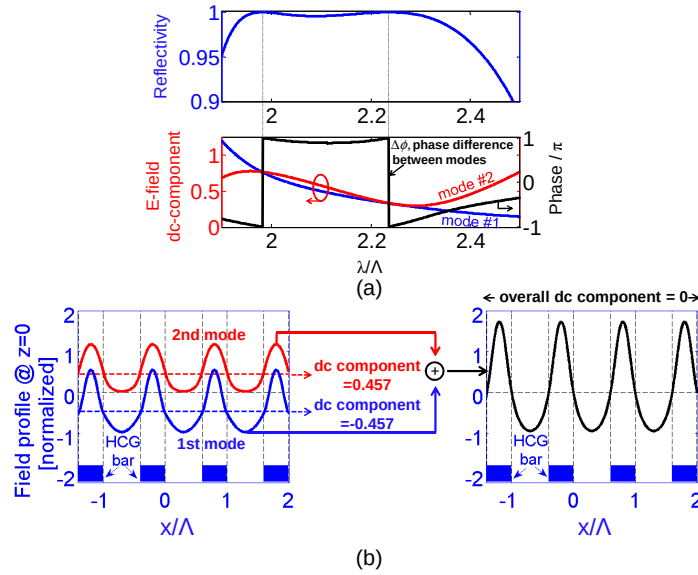


Fig. 6. (a) Double-mode solution exhibiting perfect cancellation at the HCG output plane ($z=0$) leading to 100% reflectivity. At the wavelengths of 100% reflectivity both modes have the same magnitude of the “dc” lateral Fourier component ($(a_1 + a'_1)A^{-1}e^{in_{x,1}dx} = (a_2 + a'_2)A^{-1}e^{in_{x,2}dx}$), but opposite phases: $\Delta\phi = \text{phase}[(a_1 + a'_1)A^{-1}e^{in_{x,1}dx}] - \text{phase}[(a_2 + a'_2)A^{-1}e^{in_{x,2}dx}] = \pi$. This means that the overall “dc” Fourier component is zero (Eq. (12), which leads to cancellation of the 0th transmissive diffraction order). When two perfect-cancellation points are located in close spectral vicinity of each other, a broad band of high-reflectivity is achieved. HCG parameters for this plot are: $n_{\text{bar}}=3.214$, $s/\Lambda=0.62$, $t_g/\Lambda=0.627$ and TM polarization of incidence. (b) Double-mode solution for the overall field profile at the HCG output plane ($z=0$) in the case of perfect cancellation (black curve, right plot). The cancellation is shown to be only in terms of the “dc” Fourier component. The higher Fourier components do not need to be zero, since subwavelength gratings have no diffraction orders other than 0th. The left plot shows the decomposition of the overall field profile into the first two modes, whereby the dc-components of these two modes cancel each other. HCG parameters for this plot are: $n_{\text{bar}}=3.48$, $s/\Lambda=0.4$, $t_g/\Lambda=0.2$, $\lambda/\Lambda=1.563$ and TE polarization of incidence.

4. Broadband high reflectivity mirror design

Designing a broadband HCG mirror relies on the scalability of the HCG dimensions a , s , Λ and t_g with respect to the wavelength λ . Such scalability is intuitively obvious and was reported in [1]. It is also evident from Eqs. (1)–(3). This fact greatly simplifies the design, since the first steps can be performed in normalized units: t_g/Λ , λ/Λ and $DC=s/\Lambda$, and then the normalized dimensions can be scaled according to the desired wavelength.

The first stage of the design algorithm is as follows:

- (i) The collection of solutions to Eq. (12), which are the HCG 100% reflectivity contours, are plotted on a t_g/Λ vs. λ/Λ plot. This is repeated for different grating duty cycles ($DC=s/\Lambda$), as shown in Fig. 7a. Figure 7a shows that these 100% reflectivity curves typically have an s-shape with large sections having very small slopes. Along these sections, reflectivity is 100% across a wide range of wavelengths for nearly the same value of t_g .

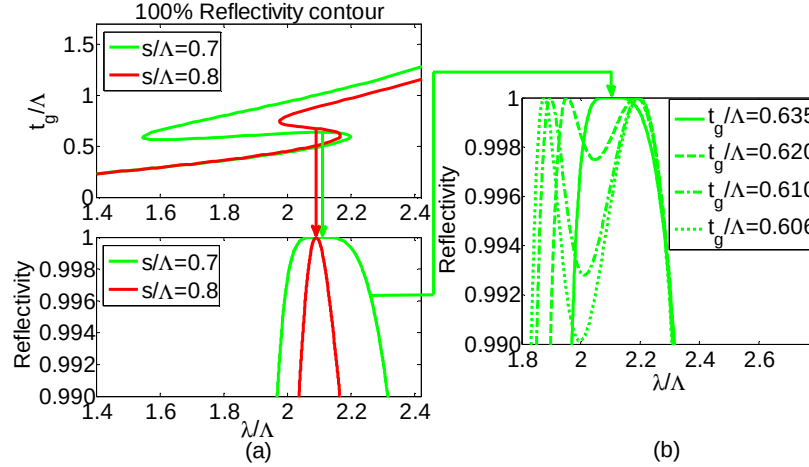


Fig. 7. (a) First stage of broadband reflectivity design: The broadband spectra are found along flat sections in 100% reflectivity contours. The 100% reflectivity contours are the solutions of Eq. (12) for different duty cycles s/Λ , marked by different colors. These contours are shown to form s-type shapes (only the lowest s-shapes are plotted). The optimal dimensions leading to the broadest spectra (indicated by arrows) are chosen. (b) Second stage of broadband reflectivity design: The bandwidth can be further increased through a tradeoff with a spectral dip, by fine-tuning the grating thickness. For example, if the minimal tolerable reflectivity for a particular application is 99%, the bandwidth can be increased by $\sim 35\%$ in comparison to a spectrally-flat configuration. Both figures (a) and (b) use normalized units, since HCG solution is scalable.

- (ii) The next step is choosing a curve section that yields the smallest slope. Such flat region would correspond to broadband high reflectivity, as described above. The arrows in Fig. 7a show examples of two choices corresponding to two different duty cycles. Among these choices, the values of $DC=s/\Lambda$, t_g/Λ and λ/Λ yielding the optimal broadest spectrum are selected.
- (iii) Finally, having selected the optimal normalized HCG dimensions, the grating period Λ is found by scaling the chosen ratio λ/Λ to fit the wavelength of interest. Having found the period Λ , the dimensions s and t_g are found using the normalized values $DC=s/\Lambda$ and t_g/Λ from the previous step.

The second stage of the design algorithm is shown in Fig. 7b: As the grating thickness, t_g , is fine-tuned, the bandwidth can be increased through a trade-off with a dip in the reflectivity spectrum. This allows maximizing the bandwidth, given the specific requirement on a minimal reflectivity that can be tolerated by a particular application. For example, if the minimal tolerable reflectivity is 99%, as shown in Fig. 7b, the bandwidth can be increased by $\sim 35\%$ in comparison to the initial spectrally-flat design, which is an output of the first design stage, described in Fig. 7a.

The *broadband* reflectivity is a result of high index contrast between the grating bars and the surrounding medium. Hence, the larger the contrast is, the wider the bandwidth. Figure 8 demonstrates this fact by plotting the same 100% reflectivity contours as in Fig. 7a for a larger grating index, $n_{\text{bar}}=4$, and for 4 different duty cycles s/Λ . The s-curve slopes in Fig. 8 are smaller and the reflectivity spectra are significantly broader. In addition, the fact that Figs. 7a, 7b and 8 use a normalized scale λ/Λ means that broadband design also automatically results in large fabrication tolerance on Λ . Large fabrication tolerance in high contrast gratings has been experimentally demonstrated by our group in the context of vertical cavity surface emitting lasers [13].

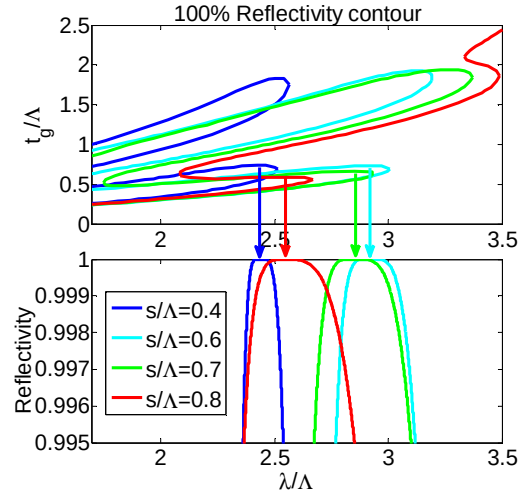


Fig. 8. Same design as in Fig. 7a repeated for a larger refractive index: $n_{\text{bar}}=4$ and four different duty cycles s/Λ . The increased refractive index contrast is shown to have a beneficial effect on the achievable high-reflectivity bandwidths.

5. Discussion – advantages and disadvantages of other solution approaches

In any electromagnetic analysis, the set of modes chosen to span the solution always has a crucial effect on both the convergence of the solution and on its physical interpretation. Usually, several different choices of mode-sets are conceivable for the same setup. A good way of determining the efficiency of the mode-set choice is its convergence, namely how many modes are required to span the solution with reasonable precision. In this work, we have chosen the mode-set of a periodic array of slab-waveguides, whereby the propagation direction is z , which is also the propagation direction of the incident plane wave. We have shown that our approach facilitates a highly efficient convergence – only two modes are required.

RCWA [9] is an example of a different mode-set choice. Its advantage is generality: RCWA can also be used for many other profiles of refractive indices, not only rectangular profiles as described in this work, and in general for many structures, other than single gratings. RCWA's disadvantage is a much slower convergence (typically on the order of ~ 10 modes, in some cases much more than 10) and a lack of physical intuition for phenomena such as high reflectivity. Another very interesting mode-set choice for similar gratings was reported recently [14], and it is based on leaky GMR (guided mode resonance) mode-set, whereby the propagation direction (within the grating) is $\pm x$. The main advantages of this approach are (i) fast convergence, requiring only 3 leaky modes (ii) harnessing a widely used terminology, such as guided mode resonances. The main disadvantages of this approach are (i) its highly qualitative nature, which lacks mathematical description, making a detailed rigorous physical analysis difficult and (ii) the fact that qualitative agreement with the GMR mode-set is only presented for TM_{0-3} modes and only for very low index-contrast gratings.

6. Conclusion

In conclusion, we have explained the nature of the $\sim 100\%$ reflectivity of high contrast dielectric gratings (HCG), classifying it as a *double-mode destructive interference* phenomenon. We presented a quickly-converging matrix transmission line formulation for the HCG reflectivity and discussed a graphic design algorithm for broadband HCG mirrors. In the context of broadband design, the high refractive index contrast proves beneficial in terms of both bandwidth and fabrication tolerance.

Acknowledgement

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